

CoRoT-2b

R-2227

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_120622 · created 2026-08-15T00:45:01+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2456100.7956 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.199 +/- 0.011
Transit depth (Rp/Rs)^2	3.95 +/- 0.43 [%]
Orbital Inclination (inc)	89.91 +/- 1.01
Airmass coefficient 1 (a1)	1.015 +/- 0.021
Airmass coefficient 2 (a2)	-0.0092 +/- 0.0085
Scatter in the residuals of the lightcurve fit is	2.05 %
Best Comparison Star	#2 - [44, 97]
Optimal Aperture	3.54
Optimal Annulus	16.25
Transit Duration (day)	0.0993 +/- 0.0011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.199 ± 0.011 vs 0.1667 (deviation 2.84σ)
Duration fit vs published	0.0993 d vs 0.095 d (deviation 3.78σ)
Mid-time O–C	-6.7 min
Depth SNR	17.5
Residual scatter	2.05 %

Light curve

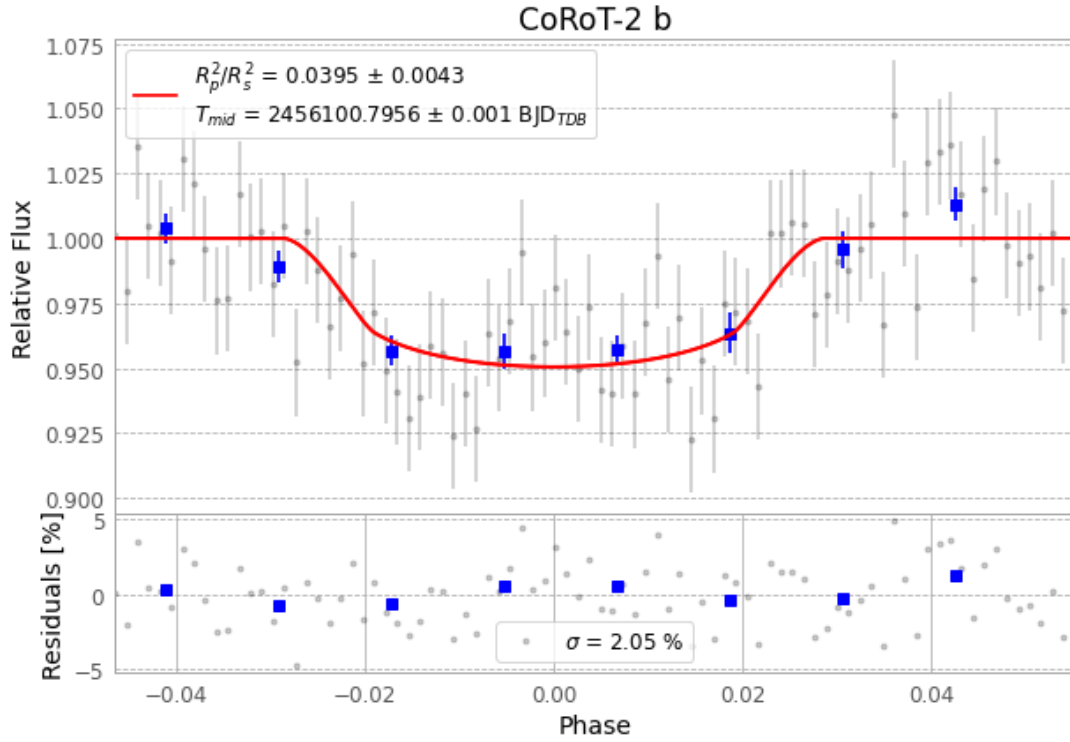


Figure 1 — FinalLightCurve_CoRoT-2 b_2012-06-21.png (SHA-256 89538affe57644b9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [303, 199], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a61eda495e3ce55f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

86 FITS frames (56 MB), CoRoT-2120622050012.FITS → CoRoT-2120622091514.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-120622.log (b68205f1d0aa...), FinalLightCurve_CoRoT-2 b_2012-06-21.png (89538affe576...), FinalLightCurve_CoRoT-2 b_2012-06-21.csv (6ecc701b8cfe...)

Compute resources

Wall time 113.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 00:45Z.